

UST'-KAMO, Russian Federation

WMO: 239920

Lat: 60.7298N

Lon: 97.5179E

Elev: 178

StdP: 99.20

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -50.1	(c) -47.0	(d) -51.8	(e) 0.0	(f) -50.0	(g) -48.9	(h) 0.0	(i) -47.1	(j) 5.1	(k) -19.3	(l) 4.4	(m) -17.9	(n) 0.0	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.8	(c) 29.2	(d) 17.9	(e) 26.8	(f) 17.0	(g) 24.6	(h) 16.2	(i) 19.2	(j) 25.8	(k) 18.2	(l) 24.6	(m) 17.2	(n) 23.1	(o) 1.7	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	12.4	21.1	15.9	11.6	20.3	14.8	10.7	19.5	55.8	25.7	52.3	24.5	49.1	23.0	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.0	(b) 5.0	(c) 4.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 6.0	(b) 5.0	(c) 4.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.6	-31.2	-25.3	-12.2	-2.9	5.0	13.6	16.6	13.1	5.7	-4.1	-18.8	-27.3		
	DBStd	17.79	10.24	8.81	8.26	5.79	4.08	4.54	3.37	3.29	4.08	6.22	10.07	10.29		
	HDD10.0	6133	1277	989	688	387	164	17	2	10	141	436	865	1156		
	HDD18.3	8749	1536	1223	946	637	414	152	72	165	380	694	1115	1414		
	CDD10.0	457	0	0	0	0	8	125	207	105	11	0	0	0		
	CDD18.3	31	0	0	0	0	0	10	19	2	0	0	0	0		
	CDH23.3	672	0	0	0	0	6	281	336	49	1	0	0	0		
	CDH26.7	172	0	0	0	0	1	82	88	2	0	0	0	0		
	CDH27.7	172	0	0	0	0	1	82	88	2	0	0	0	0		
(14) Wind	WSAvg	1.2	0.5	0.8	1.3	1.8	1.8	1.5	1.3	1.1	1.2	1.4	1.0	0.8		
(15) Precipitation	PrecAvg	496	28	21	23	31	42	51	55	62	49	55	45	34		
	PrecMax	621	55	40	45	49	77	87	97	98	83	86	88	57		
	PrecMin	349	6	7	6	15	16	10	24	30	13	13	27	9		
	PrecStd	63	11	7	9	9	14	18	21	19	16	17	15	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.0	-1.8	6.5	14.7	24.2	32.1	31.2	26.8	22.3	10.4	3.1	-1.2		
		MCWB	-5.4	-3.2	1.7	6.5	13.1	18.8	19.3	17.6	14.5	6.0	2.2	-1.7		
	2%	DB	-8.8	-4.7	4.3	11.4	20.3	29.5	29.5	24.9	17.5	7.2	-0.6	-4.7		
		MCWB	-9.5	-6.1	0.8	5.0	11.3	17.3	18.8	16.4	12.6	4.2	-1.6	-5.4		
	5%	DB	-12.2	-7.4	2.1	8.5	17.3	27.0	27.3	23.1	14.7	5.1	-2.5	-8.2		
		MCWB	-12.8	-8.3	-0.7	3.2	9.2	16.5	17.6	15.7	11.3	3.4	-3.4	-8.8		
	10%	DB	-16.1	-11.1	0.0	6.4	13.9	24.3	25.1	21.0	12.4	3.2	-4.5	-11.4		
		MCWB	-16.4	-11.8	-2.0	2.1	7.8	15.5	17.1	14.9	9.3	1.6	-5.3	-11.9		
	0.4%	WB	-5.5	-2.8	2.2	7.1	14.1	20.0	21.2	19.2	15.2	6.7	2.3	-2.2		
		MCDB	-5.1	-1.6	5.6	13.3	22.8	27.4	28.0	23.8	19.8	8.9	2.9	-1.8		
	2%	WB	-9.4	-5.8	0.9	5.3	12.1	18.6	19.9	17.8	13.4	4.7	-1.4	-5.7		
		MCDB	-8.8	-4.9	3.9	10.4	18.2	26.9	26.2	22.7	16.2	6.7	-0.7	-5.0		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	-12.7	-8.3	-0.3	3.8	10.1	17.5	18.8	16.7	11.6	3.1	-3.3	-8.9		
		MCDB	-12.2	-7.3	1.8	7.7	16.2	25.2	24.9	21.4	14.7	5.0	-2.6	-8.3		
	10%	WB	-16.4	-11.8	-1.9	2.5	8.2	16.2	17.7	15.6	9.8	1.8	-5.3	-11.9		
		MCDB	-16.0	-11.1	0.0	5.8	13.3	23.2	23.6	19.7	11.9	2.9	-4.3	-11.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	14.8	16.9	14.2	12.6	14.6	13.8	12.5	10.1	8.5	10.6	9.9		
		MCDBR	13.1	14.0	15.1	16.5	19.6	19.4	17.9	17.0	14.3	10.4	9.5	11.0		
		MCWBR	12.5	13.0	11.9	10.6	10.9	8.7	8.4	9.5	8.9	7.6	9.0	10.6		
		MCDBR	13.0	13.8	14.4	14.2	16.1	16.6	15.7	13.9	13.0	10.2	9.4	11.0		
	5% WB	MCWBR	12.5	12.8	11.5	9.2	10.1	8.2	8.1	8.5	8.5	7.5	9.0	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.300	0.328	0.361	0.377	0.403	0.452	0.433	0.350	0.314	0.256	0.232		
	taud		1.928	1.976	2.001	2.032	2.258	2.327	2.174	2.177	2.400	2.239	1.983	1.880		
	Ebn at Noon		422	645	768	820	844	826	766	744	765	646	433	295		
	Edh at Noon		48	85	117	138	119	114	130	120	80	66	44	31		
(44) All-Sky Solar Radiation	RadAvg		0.42	1.35	2.81	4.31	4.96	5.62	5.09	3.77	2.26	1.02	0.49	0.22		
	RadStd		0.04	0.10	0.20	0.36	0.39	0.56	0.41	0.27	0.34	0.14	0.07	0.03		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page